MILITARY SPECIFICATION

SHIELDS, HEAT, DISTILLING FLASK, COMPRESSED ASBESTOS AND CEMENT

1. SCOPE

1.1 **Scope.** This specification covers one type of heat shield, with centered hole, for use in distillation tests.

1.2 **Classification.** Heat shields shall be of the following sizes, as specified (see 6.1):

Size 1. - 1-1/4-inch-diameter centered hole.

Size 2. - 2-3/4-inch-diameter centered hole.

Size 3. - 4-inch-diameter centered hole.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation D256-56	Standard Methods of Test for Impact Resistance of Plastics and Electrical Insulating Materials.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material. The shields shall be composed of a combination of asbestos fiber and portland cement, formed under pressure and cured to meet the physical requirements of this specification.

3.2 Dimensions. The shields shall be square or rectangular in shape, with a centered hole, and shall conform to the nominal dimensions shown in table I.

Table I.

Size	Center hole diameter	Length	Width
	(Inches)	(Inches)	(Inches)
1	1-1/4	6	6
2	2-3/4	6	6
3	4	11	8

3.2.1 Thickness. The average thickness of the shields shall be 1/4 inch ± 1/32 inch, when tested as specified in 4.4.1.

3.3 Water absorption. The water absorption of the shields shall not exceed 22.0 percent of the dry weight, when tested as specified in 4.4.2.

3.4 Impact resistance. The impact resistance of the shields shall be 2.9 inch-pounds, when tested as specified in 4.4.3.

3.5 Clamp. A metal clamp shall be furnished with the size 3 shield for fastening the shield to a support rod.

3.6 Finish. The surfaces of the shields shall be substantially smooth.

3.7 Color. The color of the shields shall be a cement gray.

3.8 Workmanship. Workmanship shall be first class throughout. Shields shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2. For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product shall be the shield.

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Characteristic	Requirement	Test procedure	Inspection level	AQL
Thickness	3.2.1	4.4.1	S-3	*
Water absorption	3.3	4.4.2	S-3	*
Impact resistance	3.4	4.4.3	S-3	*

* Acceptance number shall be zero.

4.3 Examination. The shields shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Shield not free of cracks, tears, breaks, and deformed areas.
102 Shield warped, curled at edges, and not firmly stable on smooth flat surface.

Minor

201 Shield not free of foreign matter.
202 Shield not free of rough edges.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The shields shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Thickness. The shield shall be measured using a micrometer caliper having flat anvils approximately 1/4 inch in diameter and reading to 0.001 inch. Four measurements shall be made on each shield at the approximate midpoint of each edge of the shield and at least 1/2 inch in from the edge of the shield. The four measurements shall be averaged and recorded as the shield thickness.

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4.4.2 Water absorption. The shield shall be dried to constant weight in a ventilated oven and shall then be cooled to room temperature in a desiccator. The dry weight of the cooled shield shall be recorded and the cooled shield shall be submerged for 48 hours in clean water at 60° Fahrenheit (F.) to 80° F. The shield shall be removed from the water, wiped with a dry cloth, and weighed to determine the saturated weight. All weighings shall be made on a balance that is accurate to 0.5 percent. The water absorption value shall be calculated as follows:

$$\text{Water absorption, percent, by weight} = \frac{\text{Saturated weight - dry weight}}{\text{Dry weight}} \times 100$$

4.4.3 Impact resistance. The shield shall be tested in accordance with ASTM Designation D 256-56, method B, using the 4-inch span. The test specimen shall be unmatched and shall be cut from an area of the shield which is midway between the outer edge and the outer diameter of the centered hole.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) shield, of the size specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	(See 5.2.1.1)

5.1.3 Level A.

5.1.3.1 Unit package. Each unit (1 shield) shall be wrapped in corrugated fiberboard. Wrap shall be adequately secured to prevent accidental opening.

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5.2 Packing.

5.2.1 Level C.

5.2.1.1 Exterior container. The commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior containers. Exterior containers shall be marked with stock number, name of material, quantity, name of consignee, name of consignor, and contract or purchase order number.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Size required (see 1.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION MIL-S-36390(DM), SHIELDS, HEAT, DISTILLING FLASK, COMPRESSED ASBESTOS AND CEMENT, DATED 16 FEBRUARY 1965.

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426

WD PPSO 12502

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

FOLD UNDER

DEFENSE SUPPLY AGENCY
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

OFFICIAL BUSINESS

DEFENSE SUPPLY AGENCY

DEFENSE MEDICAL SUPPLY CENTER

ATTN: DIRECTORATE OF TECHNICAL OPERATIONS (CODE T:15)

3RD AVENUE AND 29TH STREET

BROOKLYN, N.Y., 11232

FOLD

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